

Walkthrough video — narration script

CIS4049-N Artificial Intelligence Foundations — ICA CUI JIANGKUN · F5743005

Target length $\approx$ 2 min 50 s. Speak at a relaxed pace ($\sim$ 150 words/min). Timings are cumulative. Advance the slide at each marked point.

Slide 1 — Title · (0:00-0:20)

Hello. This is my AI Foundations solution: **Autonomous Grid Navigation**. I take one problem — a warehouse robot finding its way to a drop-off — and solve it with two different AI techniques from the module, reinforcement learning and classical search, then compare them.

Slide 2 — The problem · (0:20-0:45)

The floor is a grid. The robot starts at **S** and must reach the goal **G**. Dark cells are racks it cannot cross; the yellow cells are congested aisles that cost five times as much. Crucially, the *cheapest* route is not the one with the fewest steps — and finding action sequences like this is a classic AI search problem.

Slide 3 — Two paradigms · (0:45-1:08)

Rather than pick one method, I solve it two ways. On the left, **reinforcement learning**: the robot gets no map and learns a policy by trial and error with tabular Q-learning. On the right, **classical search**: given a map, it plans the route with BFS, uniform-cost search, and A-star. Comparing them is the Week 8 “Search versus RL” question.

Slide 4 — Q-learning · (1:08-1:32)

Here is the learning. The agent acts **ϵ -greedily** — exploring early, exploiting later — and after each move applies the Q-learning update shown here. Rewards are shaped: a penalty per move, a bonus at the goal. Over about 1,500 episodes it converges — reward climbs and steps per episode fall from 200 to about 22. That plateau is a learned policy.

Slide 5 — Search algorithms · (1:32-1:54)

On the search side, three algorithms differ only in how the frontier is ordered. **BFS** uses a queue — optimal in steps, but blind to cost. **Uniform-cost search**, which is Dijkstra, orders by cost and is cost-optimal. **A-star** adds an admissible Manhattan heuristic — staying optimal while expanding far fewer nodes.

Slide 6 — Results · (1:54-2:18)

And here is the payoff. BFS takes the fewest steps but drives through the congestion for a cost of **34**. A-star and UCS both find the optimal cost of **22**, with A-star expanding just 33 nodes against 59. And **Q-learning — with no map — learns that same optimal route**. Every one of these claims is checked automatically in the notebook.

Slide 7 — Search vs RL · (2:18-2:40)

So, the trade-off. Search is exact and fast — but only because it is handed a map, and a good heuristic roughly halves the work. Reinforcement learning needs no map, which suits unknown or changing floors, but pays in training cost and gives empirical, not guaranteed, optimality. The verdict: **plan with search when the world is known; learn with RL when it is not**.

Slide 8 — Risks & close · (2:40-2:55)

Deploying this responsibly means facing the sim-to-real gap, scalability, safety certification, and workforce impact — and its limits point to future work: stochastic dynamics, SARSA, deep Q-networks, and a hybrid that plans *and* learns. Thank you for watching.

Tips: record in one take per slide if easier, then stitch. Keep the notebook or slides on screen as you narrate the matching slide. If you run long, the natural trims are the second sentences on slides 4 and 7.